Tianqi Xu

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Google Scholar | **LinkedIn** | **GitHub**

Education

Sichuan University — Chengdu, China
Bachelor of Science in Software Engineering — *Sep. 2023 – Jun. 2027 (Expected)*

- **GPA:** 3.77 / 4.0; **Weighted Average:** 89.32 / 100
- **Programs:** 985, 211, Double First Class University
- **Advisor:** Tao He

Research Interests

Vision-Language-Action (VLA) models, Test-Time Adaptation (TTA).

Publications

Tianqi Xu, Yashi Zhu, Quansong He, Yue Cao, Kaishen Wang, Zhang Yi, Tao He. **CNM-UNet: Continuous Ordinary Differential Equations for Medical Image Segmentation**, **AAAI 2026**. (Website & Paper & Code)

Research Experience

University of Central Florida — Mar. 2026 – Present
Remote Research Intern; Advisor: Yuzhang Shang — *Orlando, FL*

- Investigating the integration of **Chain-of-Thought (CoT)** reasoning into Vision-Language-Action (VLA) models to enhance embodied reasoning and long-horizon capabilities.
- Developing a **self-supervised RL** strategy for VLA post-training, enabling online trial-and-error adaptation with self-supervised rewards to enhance **generalization** without manual annotations.

Sichuan University — Feb. 2026 – Present
Onsite Research Intern; Advisor: Tao He — *Chengdu, China*

- Proposed a dynamic optimizer-state reset mechanism for test-time adaptation.
- Designed a parameter-aware thresholding strategy and a directional alignment criterion for tensor-specific anomaly detection.
- Manuscript in preparation for **ICLR 2027**.

Sichuan University — Aug. 2025 – Jan. 2026
Onsite Research Intern; Advisor: Tao He — *Chengdu, China*

- Proposed **SaCTTA**, a Continual Test-Time Adaptation method to mitigate domain shifts in medical image segmentation.
- Designed an attention-driven dynamic layer selection mechanism that treats layer outputs as information tokens to adaptively select discriminative layers.
- Introduced a self-distilled adaptive learning rate with dual constraints (sensitivity & consistency) to accelerate adaptation and suppress error accumulation.
- **Co-first author** paper submitted to **Pattern Recognition**.

Sichuan University — Sep. 2024 – Jul. 2025
Onsite Research Intern; Advisor: Tao He — *Chengdu, China*

- Proposed **CNM-UNet**, a medical image segmentation model that replaces vanilla hierarchical decoders with a single **Continuous Neural Memory ODEs** block to minimize computational complexity.
- Leveraged Neural Memory ODEs for continuous temporal feature extraction, enabling precise data distribution modeling.
- Designed a **Dual Self-updating** strategy based on test-time adaptation principles to enhance cross-domain generalization.
- **First author** paper presented at **AAAI 2026**.

Honors and Awards

- **Merit Student Award** (Consecutive 2 Years), Sichuan University 2023 – 2025
- Second-class Comprehensive Academic Scholarship (**Top 6%**), Sichuan University 2023 – 2024
- Third-class Comprehensive Academic Scholarship (**Top 15%**), Sichuan University 2024 – 2025

Skills

Languages: English (CET-4: 585, CET-6: 560), Chinese (Native).
Programming: Python, C/C++, Java, SQL.
Tools & Frameworks: PyTorch, Hugging Face, Linux, Git, Docker, OpenCV, LaTeX.